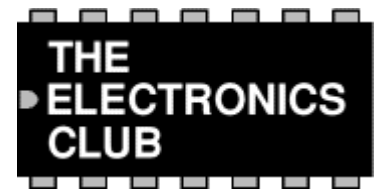


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Logic Gates



Gate types: [NOT](#) | [AND](#) | [NAND](#) | [OR](#) | [NOR](#) | [EX-OR](#) | [EX-NOR](#)
[Symbols](#) | [Truth tables](#) | [Logic ICs](#) | [Summary truth tables](#) | [Combinations](#) | [Substituting](#)

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Also see: [Logic ICs](#) | [4000 Series](#) | [74 Series](#)

Introduction

Logic gates process signals which represent **true** or **false**. Normally the positive supply voltage +Vs represents true and 0V represents false. Other terms which are used for the true and false states are shown in the table on the right. It is best to be familiar with them all.

Gates are identified by their function: [NOT](#), [AND](#), [NAND](#), [OR](#), [NOR](#), [EX-OR](#) and [EX-NOR](#). Capital letters are normally used to make it clear that the term refers to a logic gate.

Logic states	
True	False
1	0
High	Low
+Vs	0V
On	Off

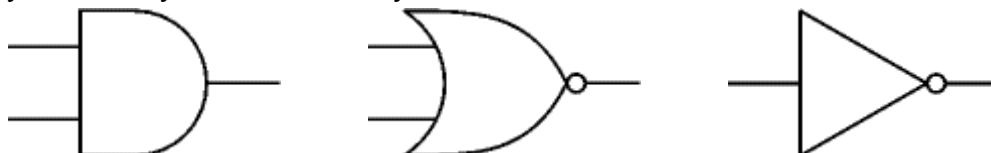
Note that logic gates are not always required because simple logic functions can be performed with switches or diodes:

- [Switches in series](#) (AND function)
- [Switches in parallel](#) (OR function)
- [Combining chip outputs with diodes](#) (OR function)

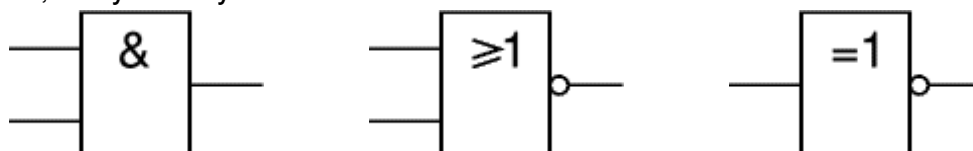
Logic gate symbols

There are two series of symbols for logic gates:

- The **traditional symbols** have distinctive shapes making them easy to recognise so they are widely used in industry and education.

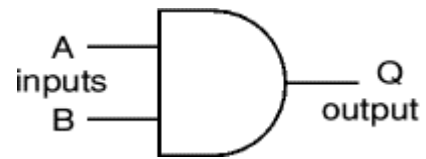


- The **IEC (International Electrotechnical Commission) symbols** are rectangles with a symbol inside to show the gate function. They are rarely used despite their official status, but you may need to know them for an examination.



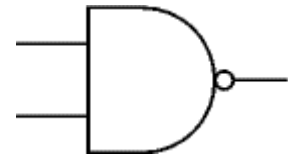
Inputs and outputs

Gates have two or more inputs, except a NOT gate which has only one input. All gates have only one output. Usually the letters A, B, C and so on are used to label inputs, and Q is used to label the output. On this page the inputs are shown on the left and the output on the right.



The inverting circle (o)

Some gate symbols have a circle on their output which means that their function includes **inverting** of the output. It is equivalent to feeding the output through a NOT gate. For example the NAND (Not AND) gate symbol shown on the right is the same as an AND gate symbol but with the addition of an inverting circle on the output.



Truth tables

A truth table is a good way to show the function of a logic gate. It shows the output states for every possible combination of input states. The symbols 0 (false) and 1 (true) are usually used in truth tables. The example truth table on the right shows the inputs and output of an AND gate.

Input A	Input B	Output Q
0	0	0
0	1	0
1	0	0
1	1	1

There are [summary truth tables](#) below showing the output states for all types of 2-input and 3-input gates. These can be helpful if you are trying to select a suitable gate.

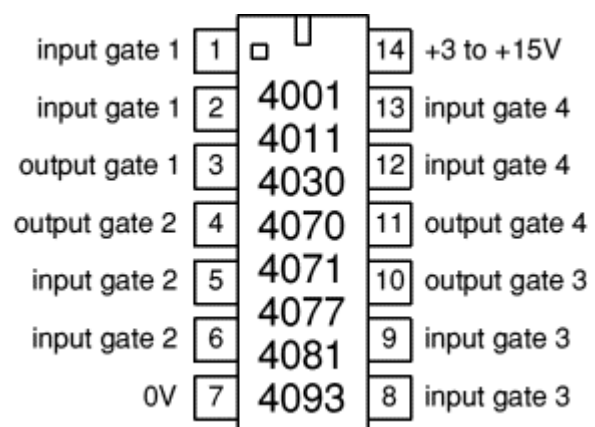
Logic ICs

Logic gates are available on special ICs (chips) which usually contain several gates of the same type, for example the 4001 IC contains four 2-input NOR gates. There are several families of logic ICs and they can be split into two groups:

- [4000 Series](#)
- [74 Series](#)

To quickly compare the different families please see:

- [Summary table of logic families](#)



The **4000** and **74HC** families are the best for battery powered projects because they will work with a good range of supply voltages and they use very little power. However, if you are using them to design circuits and investigate logic gates please remember that **all**

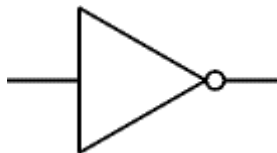
unused inputs **MUST** be connected to the power supply (either +Vs or 0V), this applies even if that part of the chip is not being used in the circuit!

NOT gate (inverter)

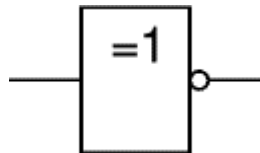
The output Q is true when the input A is NOT true, the output is the inverse of the input:

$$Q = \text{NOT } A$$

A NOT gate can only have one input. A NOT gate is also called an inverter.



Traditional symbol



IEC symbol

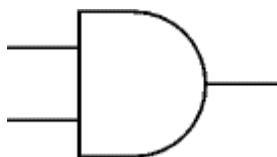
Input A	Output Q
0	1
1	0

Truth Table

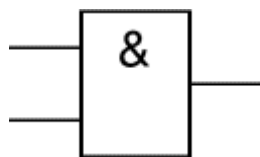
AND gate

The output Q is true if input A AND input B are both true: $Q = A \text{ AND } B$

An AND gate can have two or more inputs, its output is true if all inputs are true.



Traditional symbol



IEC symbol

Input A	Input B	Output Q
0	0	0
0	1	0
1	0	0
1	1	1

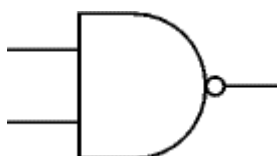
Truth Table

NAND gate (NAND = Not AND)

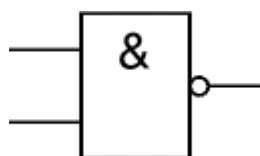
This is an AND gate with the output inverted, as shown by the 'o' on the output.

The output is true if input A AND input B are NOT both true: $Q = \text{NOT } (A \text{ AND } B)$

A NAND gate can have two or more inputs, its output is true if NOT all inputs are true.



Traditional symbol



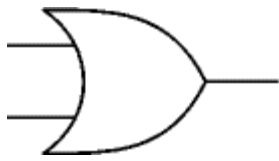
IEC symbol

Input A	Input B	Output Q
0	0	1
0	1	1
1	0	1
1	1	0

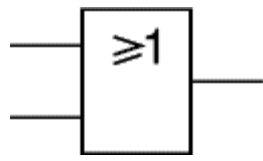
Truth Table

OR gate

The output Q is true if input A OR input B is true (or both of them are true): **$Q = A \text{ OR } B$**
 An OR gate can have two or more inputs, its output is true if at least one input is true.



Traditional symbol



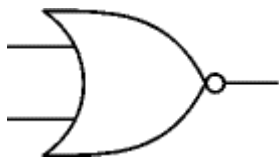
IEC symbol

Input A	Input B	Output Q
0	0	0
0	1	1
1	0	1
1	1	1

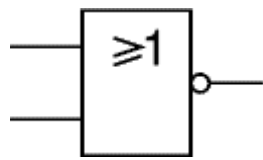
Truth Table

NOR gate (NOR = Not OR)

This is an OR gate with the output inverted, as shown by the 'o' on the output.
 The output Q is true if NOT inputs A OR B are true: **$Q = \text{NOT } (A \text{ OR } B)$**
 A NOR gate can have two or more inputs, its output is true if no inputs are true.



Traditional symbol



IEC symbol

Input A	Input B	Output Q
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table

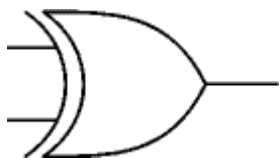
EX-OR (EXclusive-OR) gate

The output Q is true if either input A is true OR input B is true, **but not when both of them are true**: **$Q = (A \text{ AND NOT } B) \text{ OR } (B \text{ AND NOT } A)$**

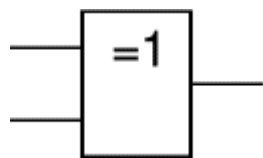
This is like an **OR** gate but excluding both inputs being true.

The output is true if inputs A and B are **DIFFERENT**.

EX-OR gates can only have 2 inputs.



Traditional symbol



IEC symbol

Input A	Input B	Output Q
0	0	0
0	1	1
1	0	1
1	1	0

Truth Table

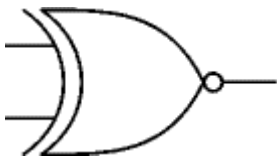
EX-NOR (EXclusive-NOR) gate

This is an EX-OR gate with the output inverted, as shown by the 'o' on the output.

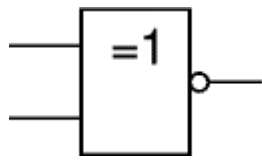
The output Q is true if inputs A and B are the **SAME** (both true or both false):

$$Q = (A \text{ AND } B) \text{ OR } (\text{NOT } A \text{ AND } \text{NOT } B)$$

EX-NOR gates can only have 2 inputs.



Traditional symbol



IEC symbol

Input A	Input B	Output Q
0	0	1
0	1	0
1	0	0
1	1	1

Truth Table

Summary truth tables

The summary truth tables below show the output states for all types of 2-input and 3-input gates.

Summary for all 2-input gates							
Inputs		Output of each gate					
A	B	AND	NAND	OR	NOR	EX-OR	EX-NOR
0	0	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	0	1	0	0	1

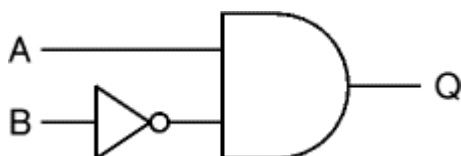
Note that EX-OR and EX-NOR gates can only have 2 inputs.

Summary for all 3-input gates						
Inputs			Output of each gate			
A	B	C	AND	NAND	OR	NOR
0	0	0	0	1	0	1
0	0	1	0	1	1	0
0	1	0	0	1	1	0
0	1	1	0	1	1	0
1	0	0	0	1	1	0
1	0	1	0	1	1	0
1	1	0	0	1	1	0
1	1	1	1	0	1	0

Combinations of logic gates

Logic gates can be combined to produce more complex functions. They can also be combined to **substitute** one type of gate for another.

For example to produce an output Q which is true only when input A is true and input B is false, as shown in the truth table on the right, we can combine a NOT gate and an AND gate like this:



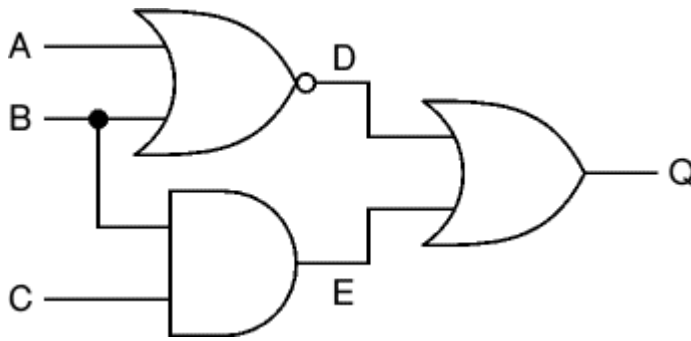
Input A	Input B	Output Q
0	0	0
0	1	0
1	0	1
1	1	0

$$Q = A \text{ AND NOT } B$$

Working out the function of a combination of gates

Truth tables can be used to work out the function of a combination of gates.

For example the truth table on the right show the intermediate outputs D and E as well as the final output Q for the system shown below.



Inputs			Outputs		
A	B	C	D	E	Q
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	0	1	1

$$D = \text{NOT } (A \text{ OR } B)$$

$$E = B \text{ AND } C$$

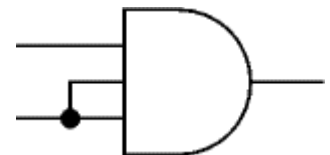
$$Q = D \text{ OR } E = (\text{NOT } (A \text{ OR } B)) \text{ OR } (B \text{ AND } C)$$

Substituting one type of gate for another

Logic gates are available on ICs which usually contain several gates of the same type, for example four 2-input NAND gates or three 3-input NAND gates. This can be wasteful if only a few gates are required unless they are all the same type. To avoid using too many ICs you can reduce the number of gate inputs or substitute one type of gate for another.

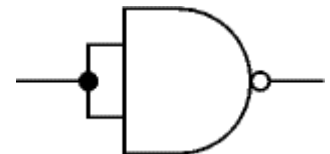
Reducing the number of inputs

The number of inputs to a gate can be reduced by connecting two (or more) inputs together. The diagram shows a 3-input AND gate operating as a 2-input AND gate.



Making a NOT gate from a NAND or NOR gate

Reducing a NAND or NOR gate to just one input creates a NOT gate. The diagram shows this for a 2-input NAND gate.



Any gate can be built from NAND or NOR gates

As well as making a NOT gate, NAND or NOR gates can be combined to create any type of gate! This enables a circuit to be built from just one type of gate, either NAND or NOR. For example an AND gate is a NAND gate then a NOT gate (to undo the inverting function). Note that AND and OR gates cannot be used to create other gates because they lack the inverting (NOT) function.

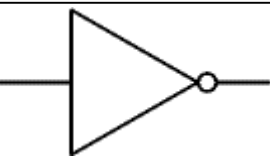
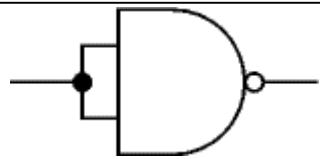
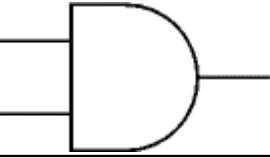
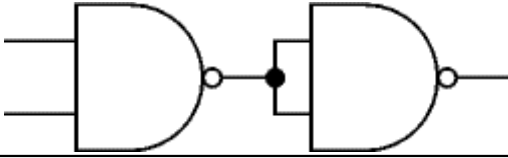
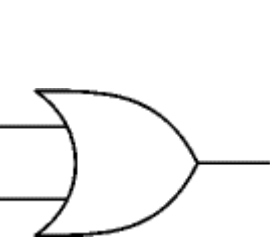
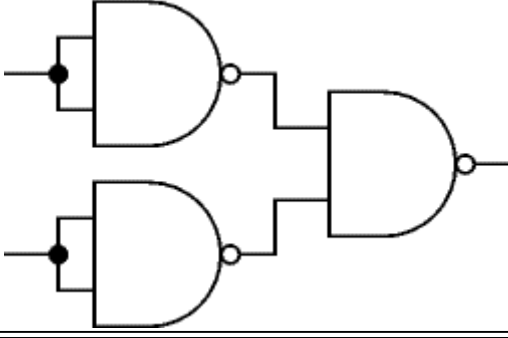
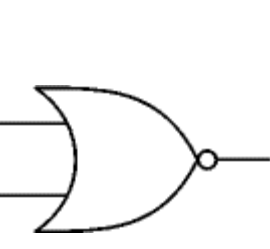
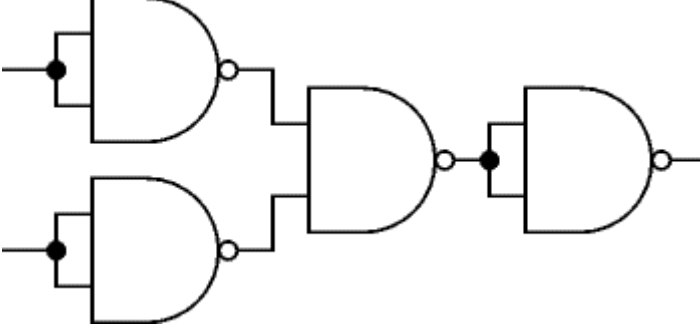
To change the type of gate, such as changing OR to AND, you must do three things:

- Invert (NOT) each input.
- Change the gate type (OR to AND, or AND to OR)
- Invert (NOT) the output.

For example an OR gate can be built from NOTed inputs fed into a NAND (AND + NOT) gate.

NAND gate equivalents

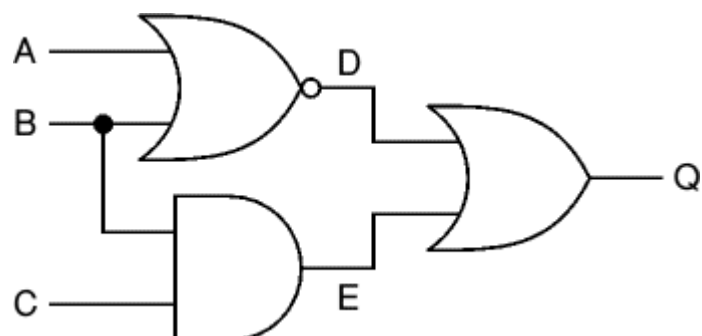
The table below shows the NAND gate equivalents of NOT, AND, OR and NOR gates:

Gate		Equivalent in NAND gates
NOT		
AND		
OR		
NOR		

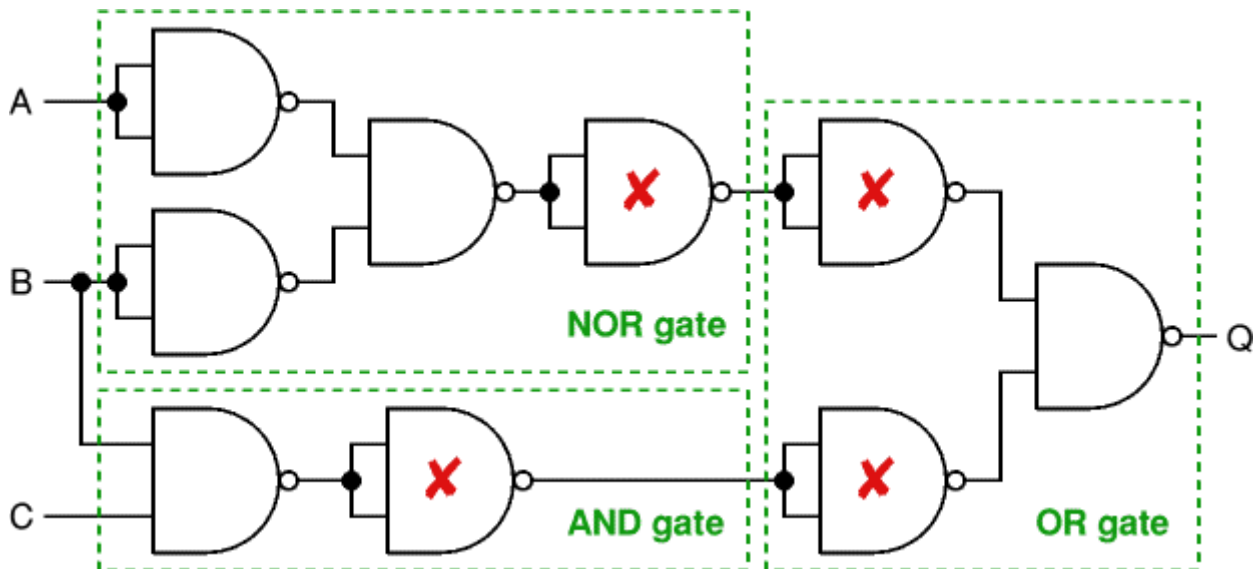
Substituting gates in an example logic system

The original system has 3 different gates: NOR, AND and OR. This requires three ICs (one for each type of gate).

To re-design this system using NAND gates only begin by replacing each gate with its NAND gate equivalent, as shown in the diagram

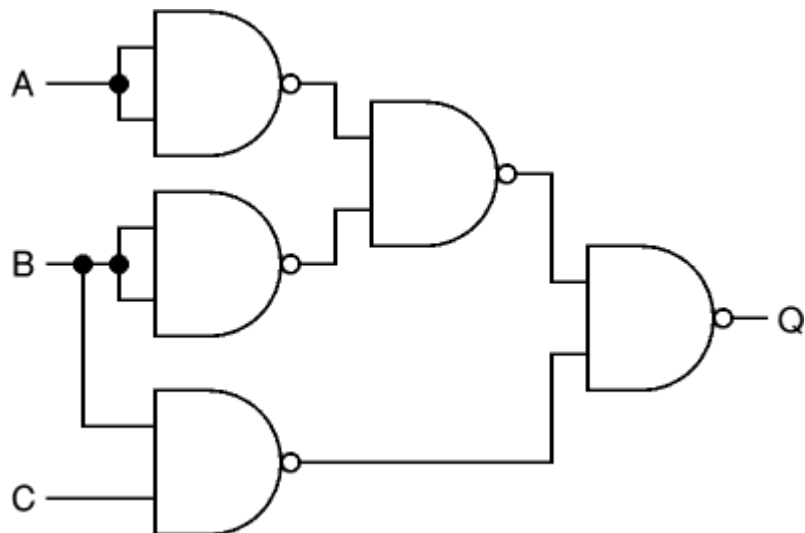


below.



Then simplify the system by deleting adjacent pairs of NOT gates (marked **X** above). This can be done because the second NOT gate cancels the action of the first.

The final system is shown on the right. It has five NAND gates and requires two ICs (with four gates on each IC). This is better than the original system which required three ICs (one for each type of gate).



Substituting NAND (or NOR) gates does not always increase the number of gates, but when it does (as in this example) the increase is usually only one or two gates. The real benefit is reducing the number of ICs required by using just one type of gate.

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